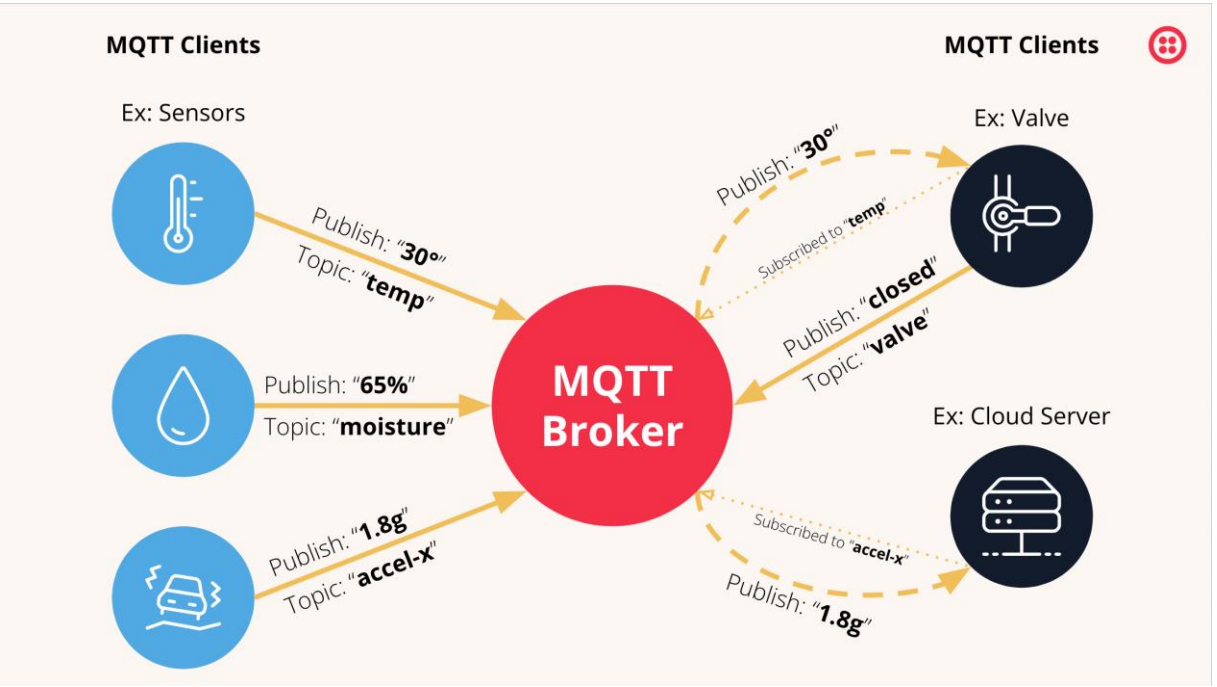
ICT30016 IoT Track – Week 6

IoT Communications with MQTT

Prepared by Dr. Mark Tee Kit Tsun (SUTS)

What is MQTT?

- Message Queuing Telemetry Transport (MQTT) is a data exchange Protocol (a way to pass messages).
- Especially used for Internet-of-Things applications because it is capable for massive Many-To-Many communications.
- Uses a Publish-Subscribe type of Client-Server Architecture.
  - Many receivers Subscribe to a Topic
  - When one sender publishes a message to that Topic, all those receivers get a copy of it



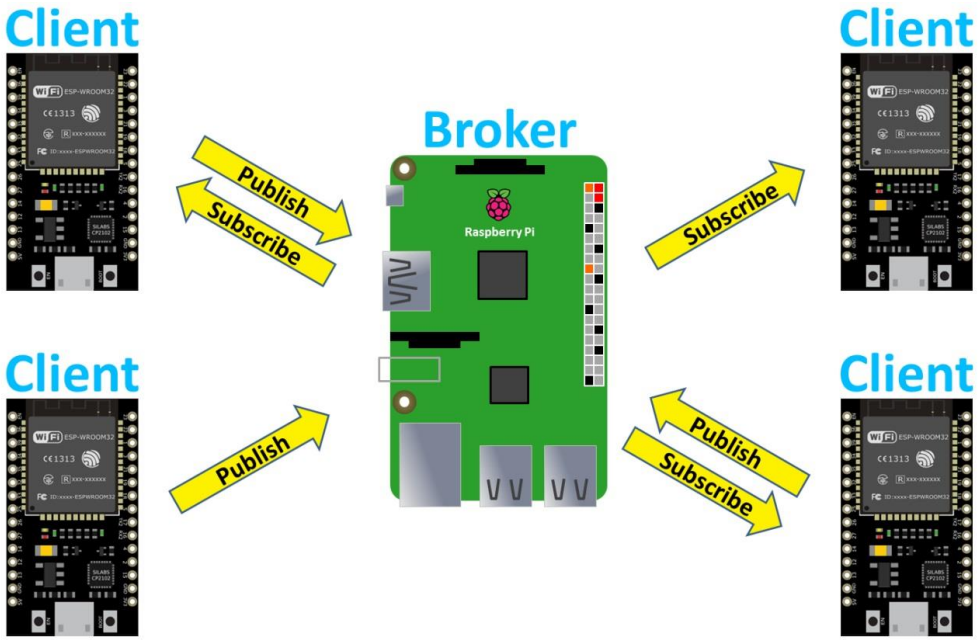


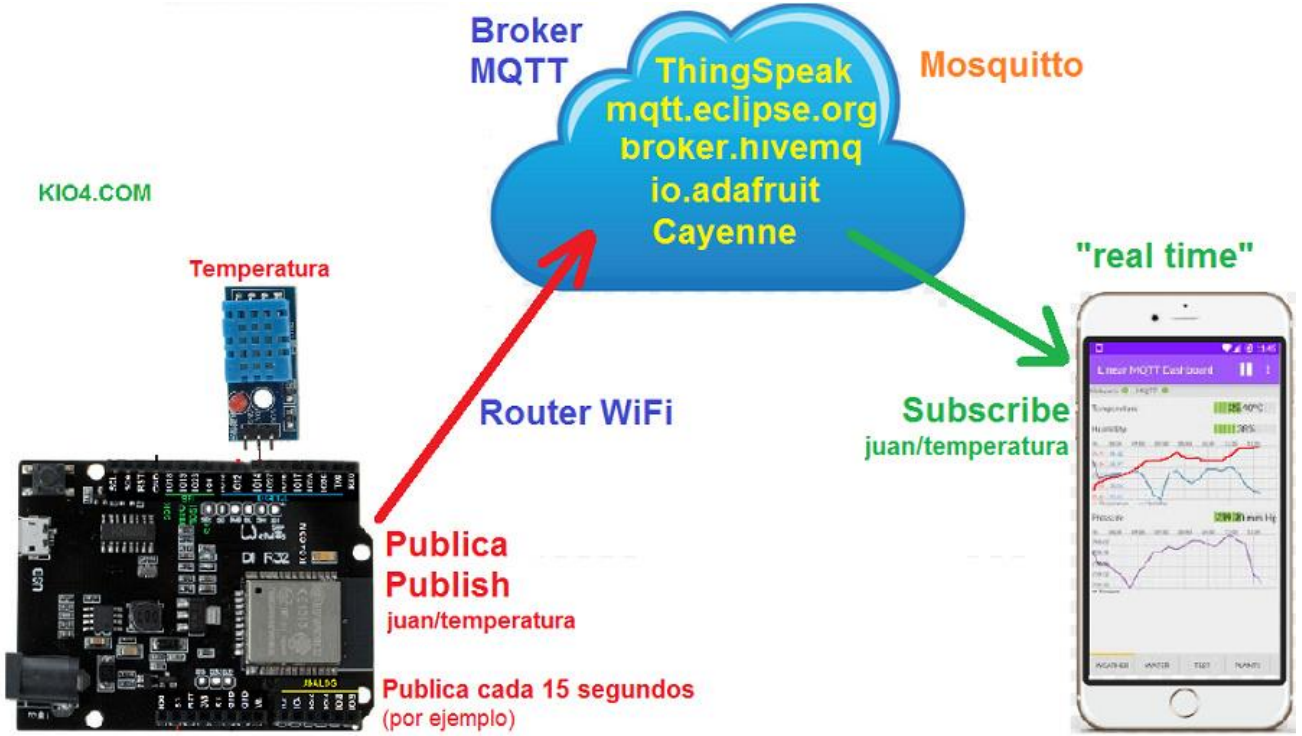
MQTT Basics: What is MQTT and How Does it Work? - YouTube

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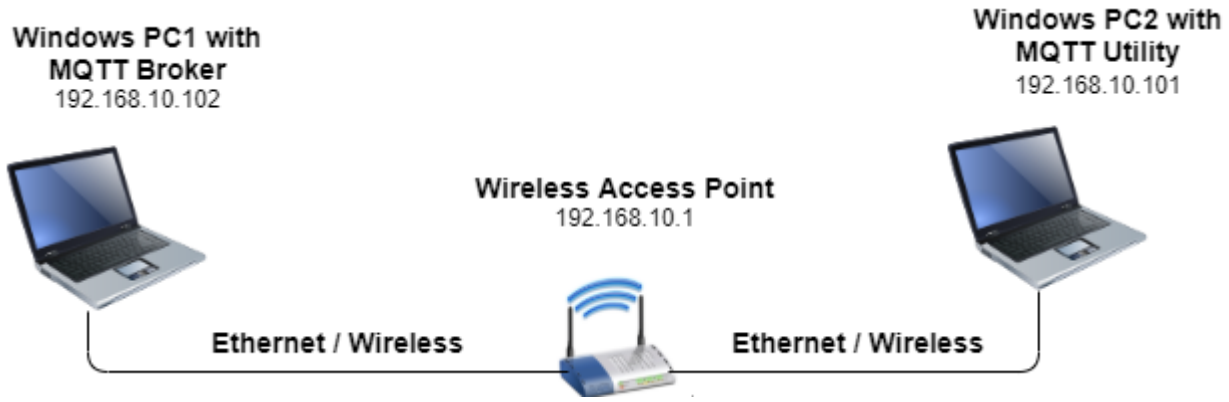
Why should I care?

- MQTT allows you to send and receive messages from multiple ESP32 machines, PCs, smartphones and more.
- You just need to download, install and configure an MQTT Broker in a computer on the same network.
- Each of the MQTT “Clients” must have a PubSubClient library to send and receive messages.
- There is an Open Source (AKA Free) MQTT Broker you can use, called Mosquitto.





Task 1: Setup a Mosquitto MQTT Broker on your PC



- The Mosquitto MQTT Broker is a server program that runs in the background of your computer.
- Follow the following tutorial to:
  - Download and Install Mosquitto on your Windows Computer
  - Configure the mosquitto.conf configuration file to allow anonymous mode and set your port to 1883.
  - Do Settings on your Firewall to allow Mosquitto to operate.

https://shop.theengs.io/blogs/news/installing-mosquitto-on-windows-and-make-it-accessible-from-your-local-network

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Task 2: Program the ESP32 to Connect to MQTT Broker

- You need to download and include the PubSubClient library in your Arduino IDE.
- Then, follow the tutorial to implement code in your NodeMCU ESP32 to practice sending and receiving messages via MQTT.
- You must have a working MQTT Broker running in a PC on the same WIFI first. Tutorial:

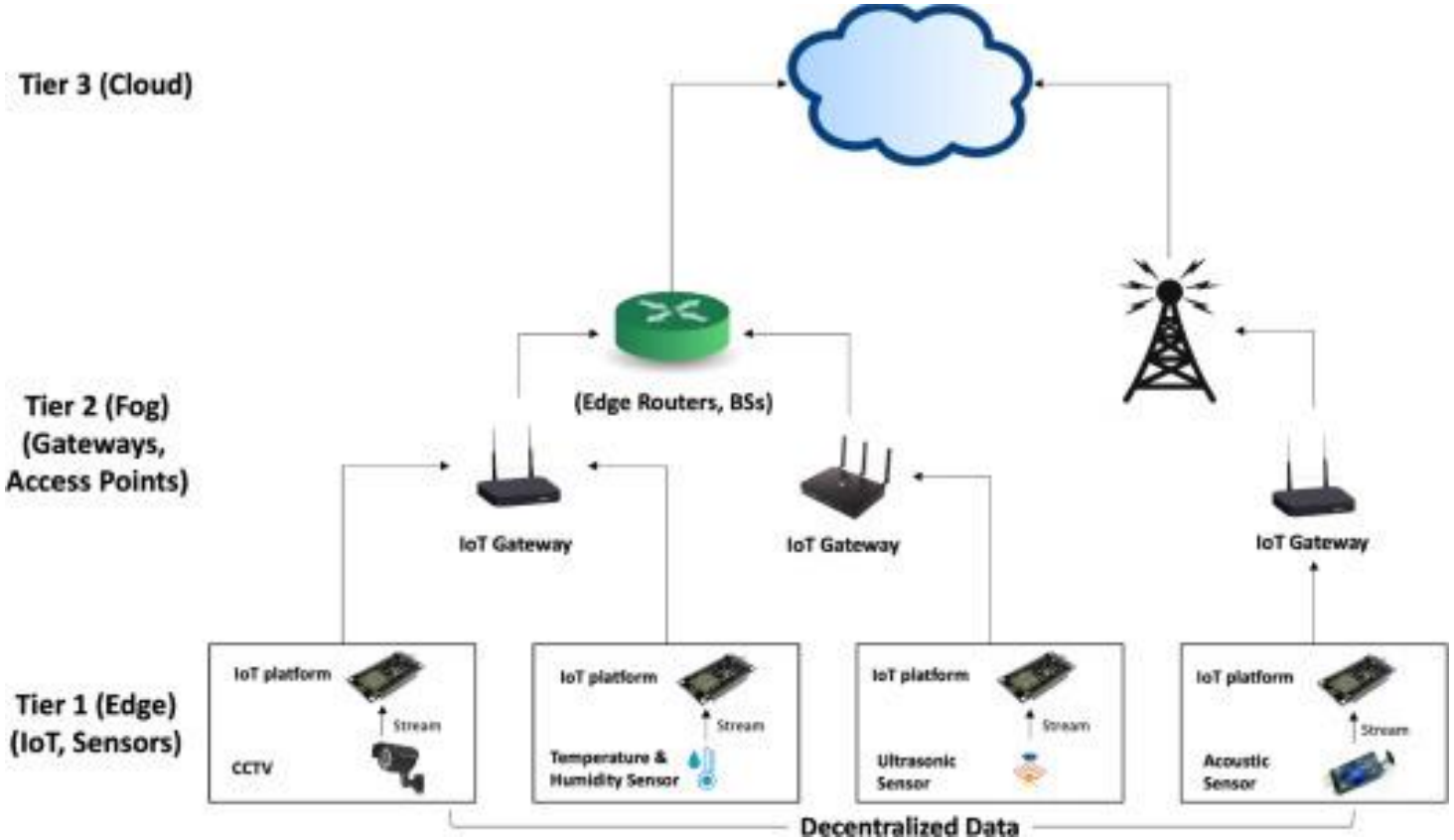


https://www.youtube.com/watch?v=7mYNaVNoIg0

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Challenge: Node-to-Node Communications

- The heart of an IoT system is the ability for Nodes to send messages to one another.
- Each Device or Computer that participates in the IoT system is called a Node.
  - Example: ESP32, Raspberry Pi, Laptop/PC, Virtual Machine on EC2 in AWS, smartphone, etc.
- Each Node must be connected to the other via some form of Network. This can be visualized in the form of a multi-tiered Network Diagram (part of Week 1 topic).



Use MQTT to facilitate this messaging system so your ESP32 sends and receives messages to your PC as part of your Project.

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Takeaway for this Week

1. The MQTT Broker is the heart of most IoT system in the world – allowing IoT devices to send messages to each other across the network.

2. Mosquitto is a popular, open source MQTT Broker, but there are a lot of alternatives such as HiveMQ – an online broker service, that might be easier to try with.

3. There's a large number of tutorials concerning "ESP32 with MQTT and Arduino" so search online to see which ones work best for you. E.g.:
   - MQTT Tutorial using Arduino Framework (donskytech.com)
   - HOW TO CONTROL LED WITH PUBLISH/SUBSCRIBE MQTT IOT (youtube.com)
   - ESP32 MQTT Publish Subscribe with Arduino IDE | Random Nerd Tutorials
   - MQTT on ESP32: A Beginner's Guide | EMQ (emqx.com)

Other PCs can also participate as a MQTT client. Again, look online for similar tutorials:

- Implementing MQTT in Python (hivemq.com)
- How to Use MQTT in C# with MQTTnet | EMQ (emqx.com)

Phone Apps are also available to quickly test MQTT messaging:
Android and MQTT: A Simple Guide. How to develop an MQTT Client with… | by Leonardo Cavagnis | The Startup | Medium

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